



Introduction to

Agentic AI

Foundations & Real-World Understanding



Unit 1 Roadmap

Day 1

Session 1

Foundation of Agentic AI

AI Agents · Gen AI vs Agentic AI

Day 2

Session 2

Use Cases & Agent Types

Industry Use Cases · Types · Future Trends

Day 3

Session 3

Core components: Model, Tools & Memory

Model · Tools · Memory · ReAct · ReWOO

Day 4

Session 4

Agentic architectures: ReAct & ReWOO + Setup labs

Ollama · Python · IBM Cloud API

Learning Objectives

By end of this session →

Agentic AI · Day 1

01 Define AI and explain how it learns, reasons, and decides

02 Trace AI evolution from Rule-Based → Agentic AI across 5 stages

03 Explain how RAG works and its capability boundaries

04 Describe Agentic AI's perceive → plan → act → iterate loop

05 Compare RAG vs Agentic AI using real college scenarios

06 Name the 5 core components of every AI agent

07 Classify AI systems across 4 levels of autonomy

08 Trace the AI decision cycle: Observe → Analyze → Plan → Act → Learn

09 Assess AI risks and explain who is accountable for AI mistakes

10 Distinguish Generative AI from Agentic AI by memory, tools & output

Are These Tools...

or Something More?



Siri

Answers your
questions



Alexa

Controls your smart
home



ChatGPT

Generates text
responses



Google Maps

Plans your route

Are they just tools, or intelligent assistants?

Yes — These Are Early AI Agents



Siri

Answers your questions



Alexa

Controls your smart home



They assist — understand your intent



ChatGPT

Generates text responses



Google Maps

Plans your route



They predict — anticipate your needs



They guide — help make decisions

This is the beginning of Agentic AI

AI: Machines That Think Like Humans



Learn

Improves from experience



Reason

Analyzes & draws conclusions



Decide

Chooses the best action

Real-World Examples



Chatbots

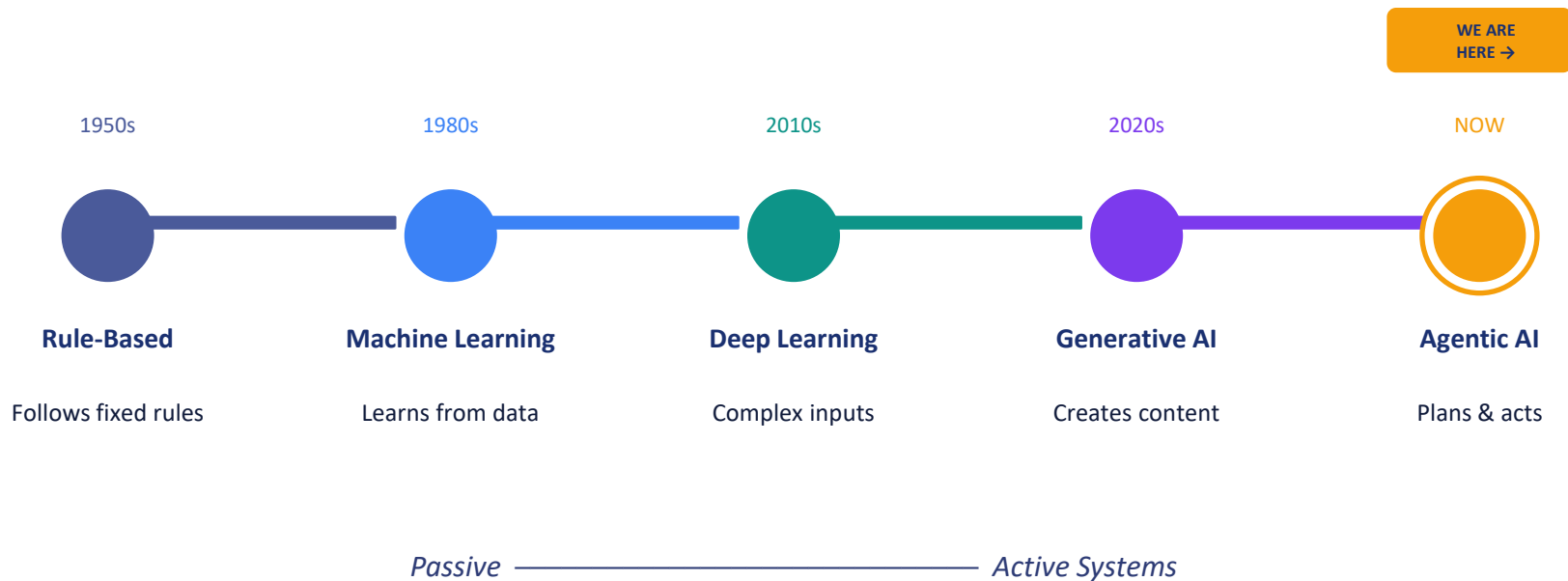


Recommendation
Engines



Self-Driving Cars

AI Has Evolved — Step by Step



"Every evolution gave AI more power. Agentic AI gives it autonomy."



Quick Quiz — Match the AI Stage



Attendance System



?



Result Prediction



?



Face Unlock



?



ChatGPT



?



Smart Assistants



?

Stage Labels:

Machine Learning

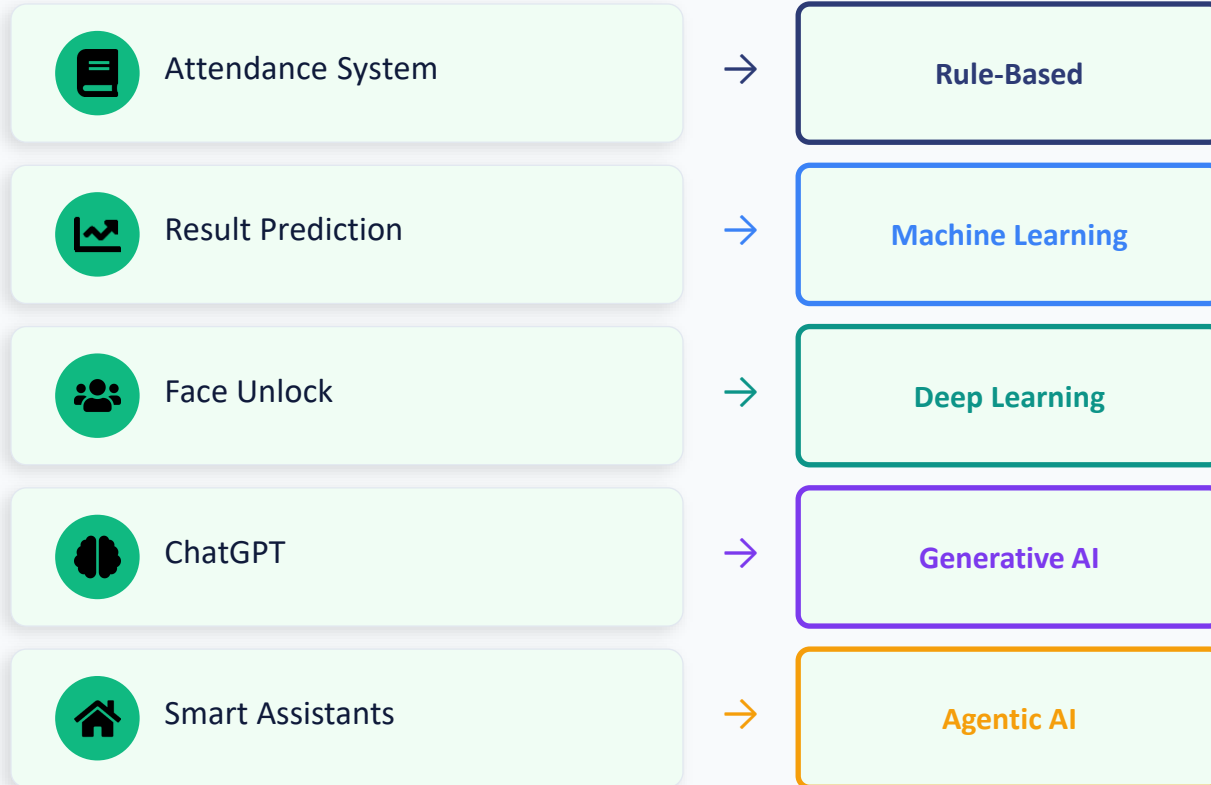
Generative AI

Deep Learning

Agentic AI

Rule-Based

Here's How They Map



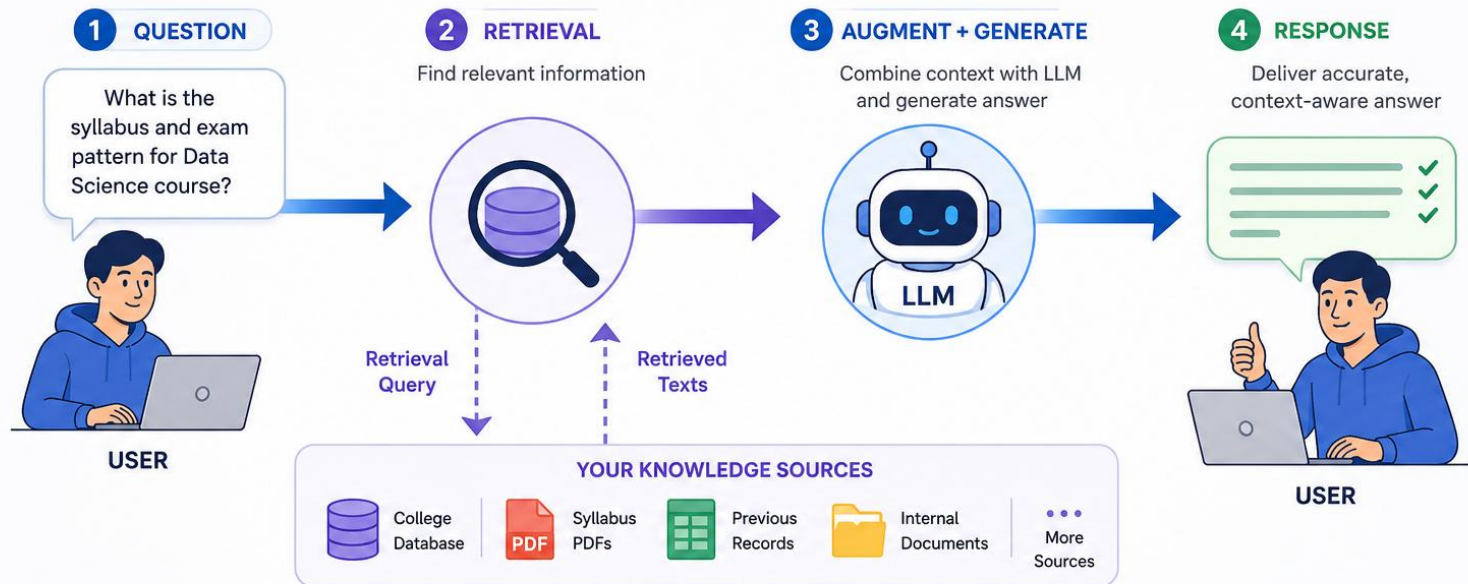
Let's See AI in Action

Two systems. One college scenario.

Which one can really help?

Retrieval Augmented Generation

Better Answers with Your Data



WHY RAG?



Context-aware answers



Uses your trusted data



More accurate, less hallucination



Does NOT take actions

RAG Chatbot: Answering Your Questions

Goal: "Tell me the syllabus and exam pattern for Data Science"



Retrieve Syllabus

From college database



Fetch Exam Pattern

From previous records



LLM Processes

Combines & understands



Response Generated

Clear answer delivered

 **DataBot — College Assistant**

Tell me the syllabus for Data Science

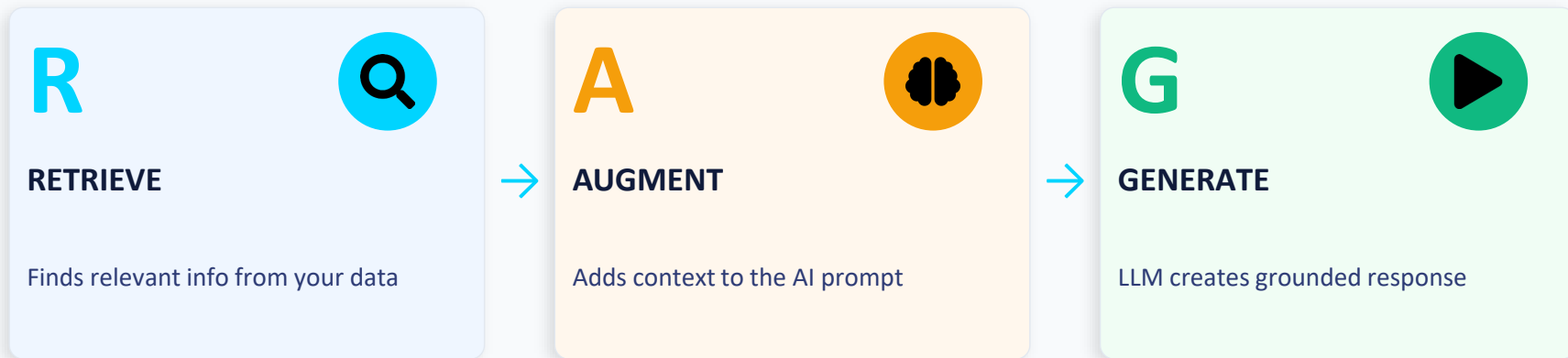
Here's the Data Science syllabus:

- Module 1: Python Basics
- Module 2: Statistics
- Module 3: ML Algorithm

Exam: 70% written + 30% practicals

RAG makes AI knowledgeable — but not capable of action.

RAG = Retrieval + Augmentation + Generation



What RAG Can vs Cannot Do

✓ Can Do

- Answer context-based questions
- Retrieve from knowledge base
- Reduce AI hallucination

✗ Cannot Do

- Send emails or take actions
- Create student profiles
- Execute multi-step tasks

What RAG Can (and Cannot) Do



RAG Can

Answer context-based questions

Retrieve from your knowledge base

Reduce AI hallucination

Cite specific documents



RAG Cannot

Send emails or messages

Create student profiles

Execute multi-step tasks

Use external tools/APIs

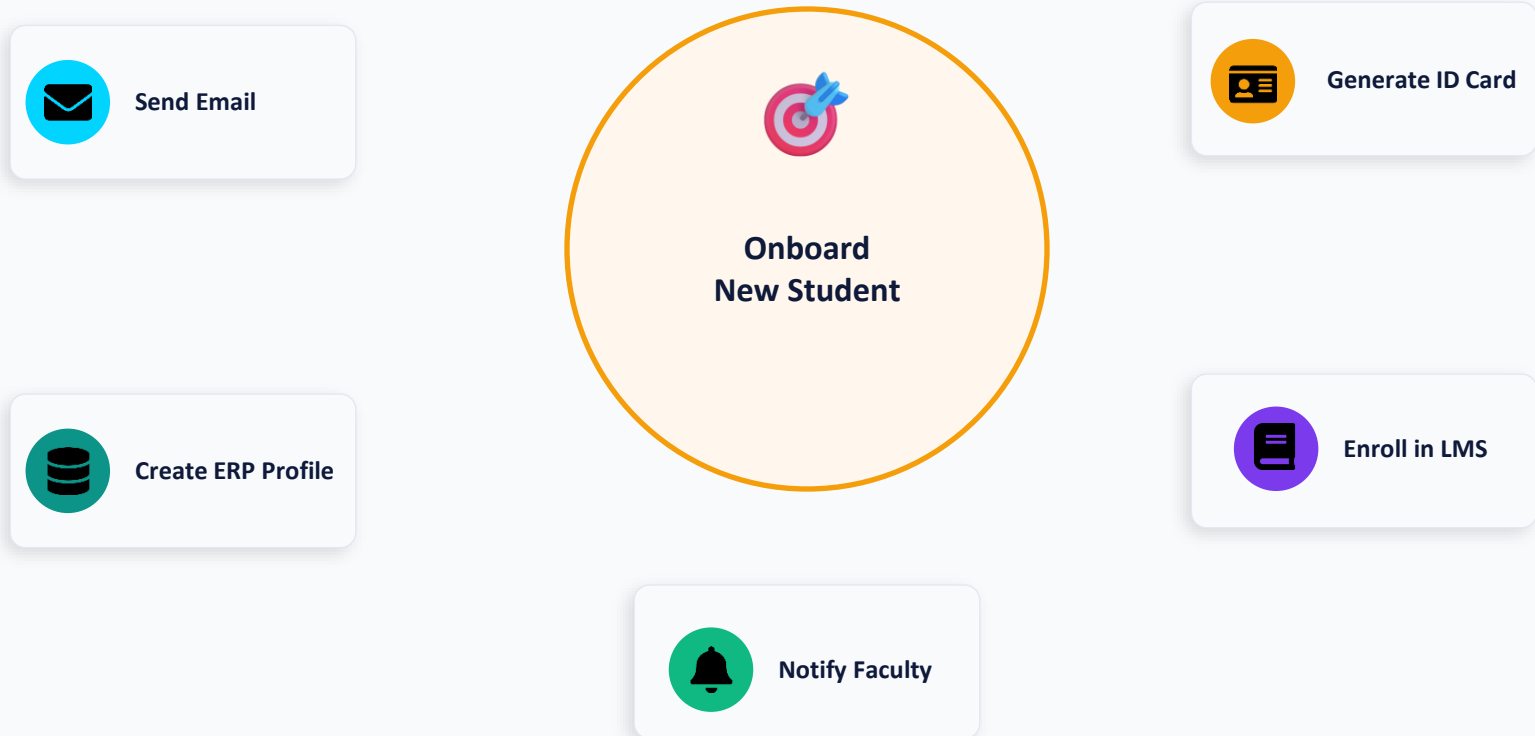
"RAG is a smart librarian — not an executive assistant"

What If AI Could Actually **DO It?**

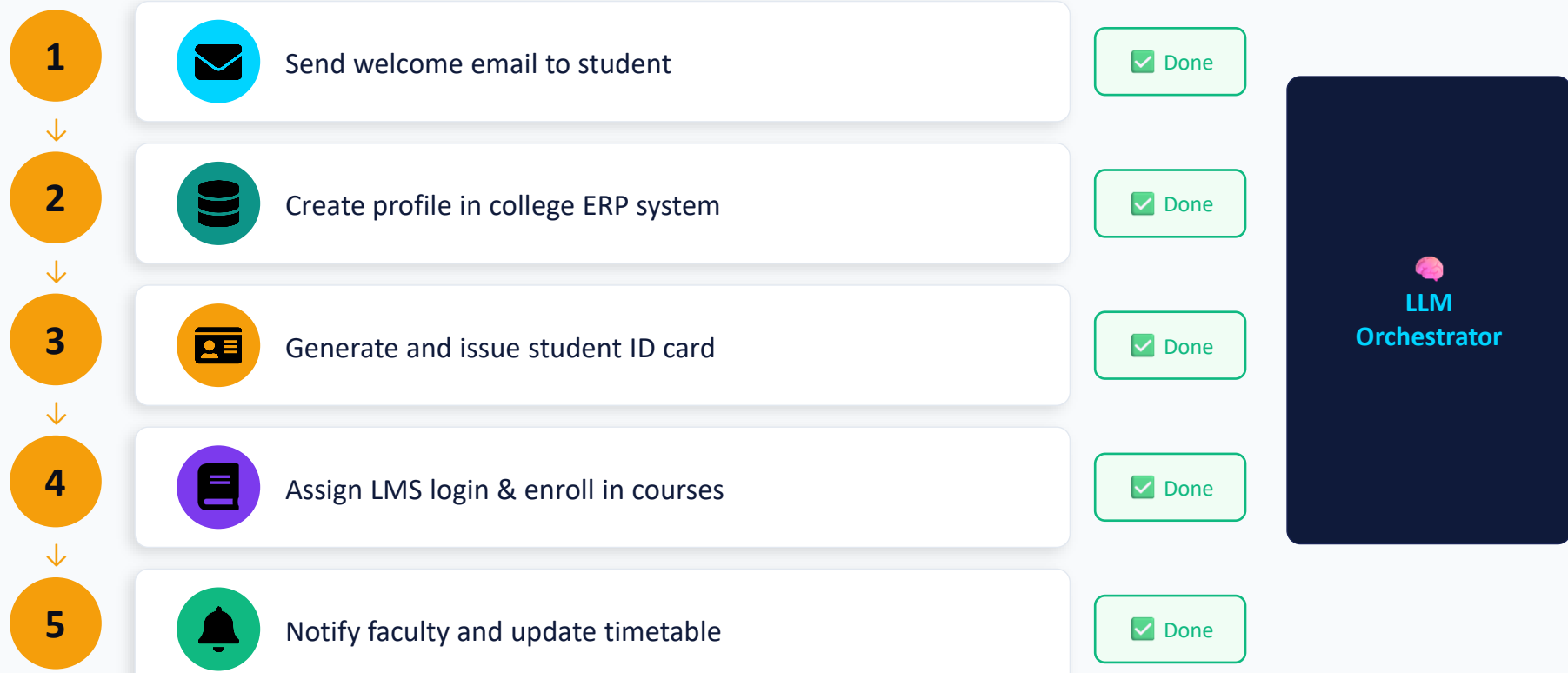
Not just answer — but complete the entire task?

Agentic AI: Onboarding a New Student

One instruction → complete multi-step workflow



Watch the Agent Work — Step by Step



One goal, five tools, zero human intervention — that's Agentic AI.

Agentic AI = Think + Plan + Act

Think

Understands your goal

Plan

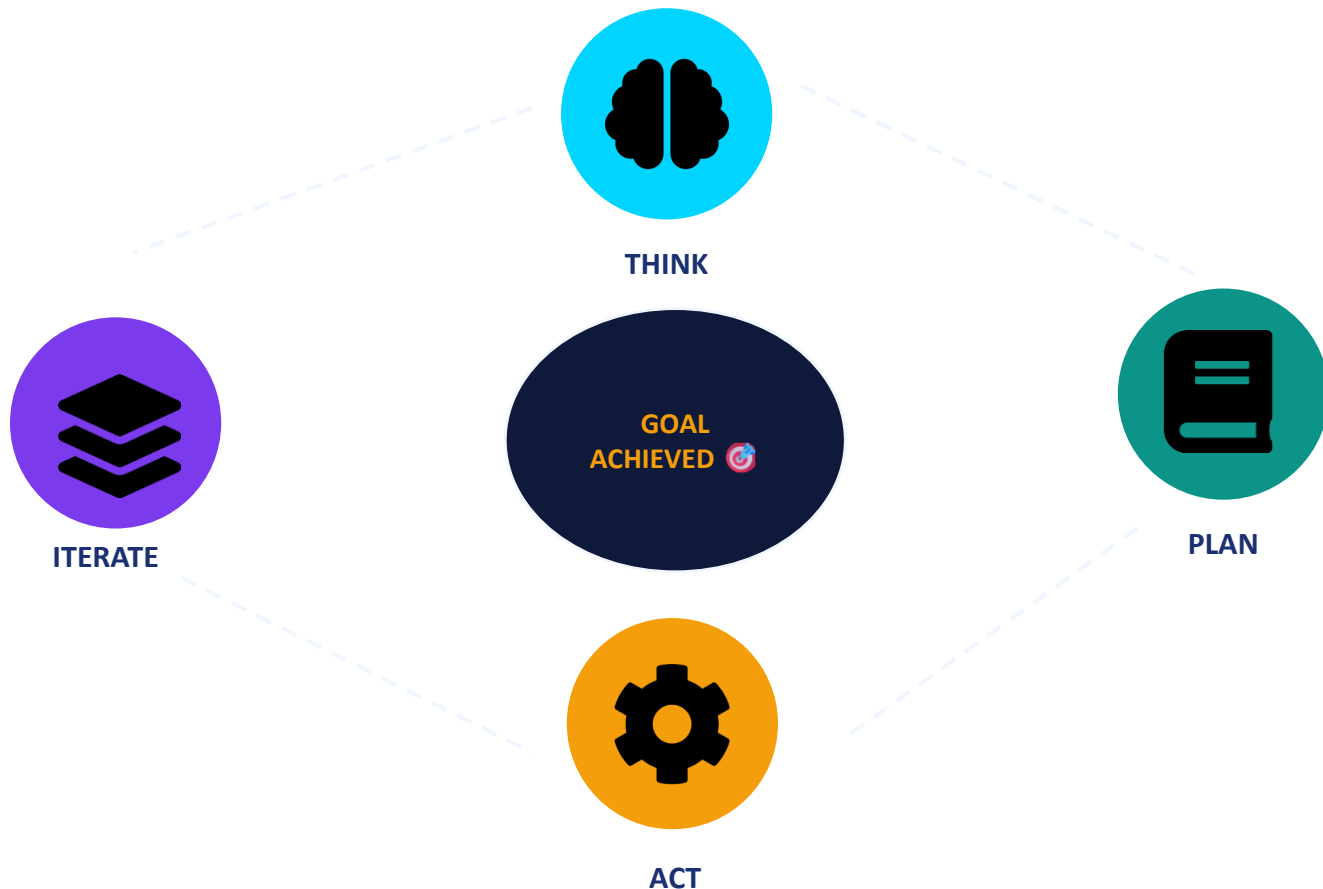
Breaks into ordered steps

Act

Executes using real tools

Iterate

Adjusts if something fails



Agentic AI: A New Level of Power



Goal-Oriented Planning

Breaks complex goals into ordered tasks



Multi-Step Reasoning

Thinks ahead through dependencies



Autonomous Decisions

Acts without constant human input



Tool Usage

Uses ERP, Email, APIs, Code



Memory & Context

Retains context across sessions

Let's Compare the Two

RAG vs Agentic AI



RAG vs Agentic AI — Side by Side

Feature	 RAG Chatbot	 Agentic AI
Purpose	Answer questions	Complete tasks
Data Usage	Retrieves info	Uses tools + data
Takes Actions	 No	 Yes
Memory	Limited (session)	Persistent
Example	"What is syllabus?"	"Enroll me in AI course"

Which one can complete your full request?

Which System Gets the Job Done?

"Register me for the AI course and send me a confirmation email"

RAG Chatbot

✗ Sorry, I can only answer
'What is the AI course?'
I cannot register you.

Can only answer questions.

Agentic AI

✓ Registered for AI Course

✓ Confirmation email sent

✓ Task complete!

Completes the full task.

Brain vs Brain + Hands



 **RAG**

Answers

Knowledge Only

VS



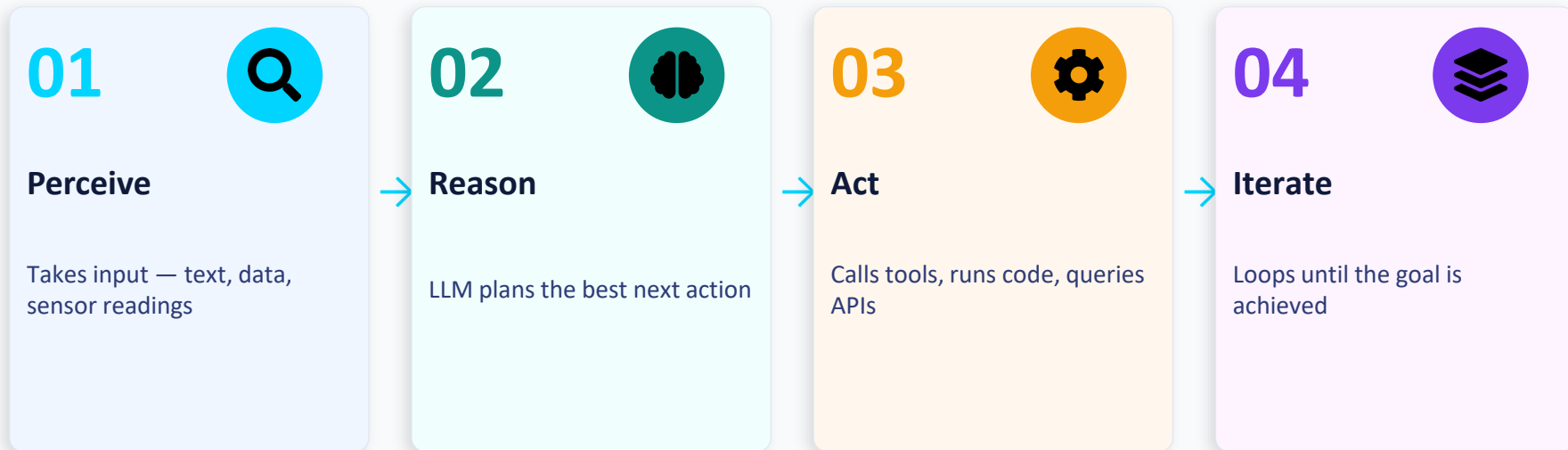
  **Agentic AI**

Acts

Knowledge + Action

"The future belongs to AI that doesn't just know — but does."

AI Agent: The 4-Step Definition



“ An AI agent is software that perceives, reasons, acts, and iterates — autonomously. ”

5 Core Components of Every AI Agent



Perception

Collects info from environment



Reasoning

Analyzes and interprets data



Decision-Making

Selects the best action



Action

Executes the chosen task



Learning

Improves from feedback

What Makes Agentic AI Different?

BEHAVIORAL TRAITS



Autonomy



Goal-Oriented



Reactive



Proactive



Adaptive

TECHNICAL CAPABILITIES



Tool Use



Memory



Self-Correction



Multi-Step
Reasoning

Which Characteristic Is This?

"A chatbot improves its responses based on past conversations"

A) Reactive

B) Adaptive

C) Autonomous

Choose the best answer →

Answer: Adaptive

"A chatbot improves its responses based on past conversations"

A) Reactive

B) Adaptive

C) Autonomous

Adaptive = learns from history and improves over time

4 Levels of AI Autonomy

Agentic AI Target

1

Human Controlled

You do everything



e.g. Calculator

2

Human-in-the-Loop

AI suggests, human approves



e.g. Medical AI

3

Semi-Autonomous

AI acts, human oversees



e.g. Smart Drone

4

Fully Autonomous

AI operates independently



e.g. Self-driving Car

Rank from Least to Most Autonomous

Place these in order (low → high autonomy):



GPS Navigation



Smart Vacuum

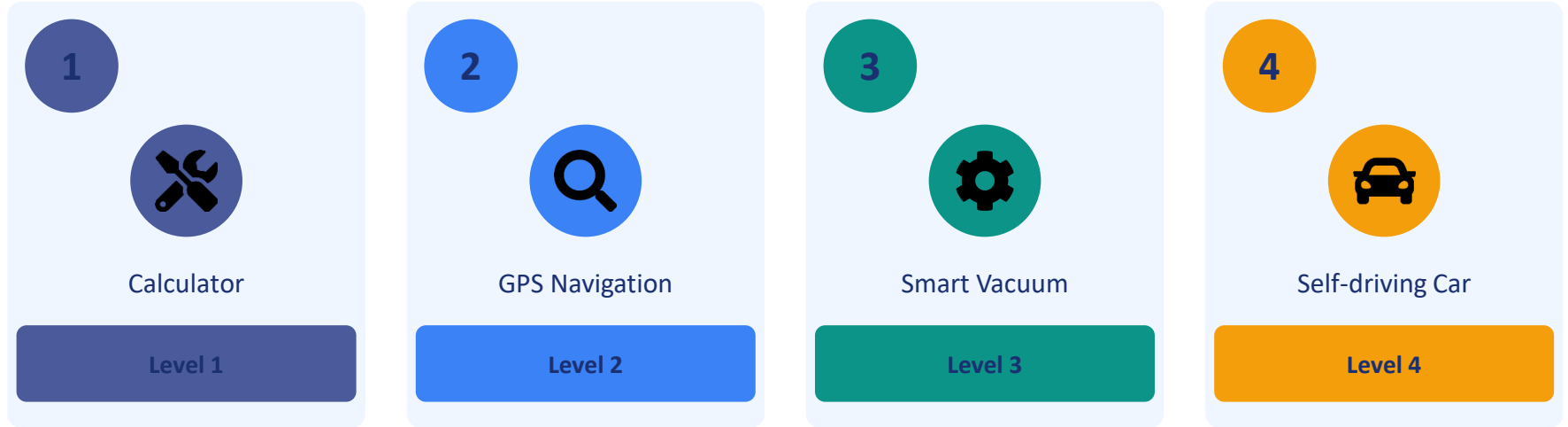


Calculator



Self-driving Car

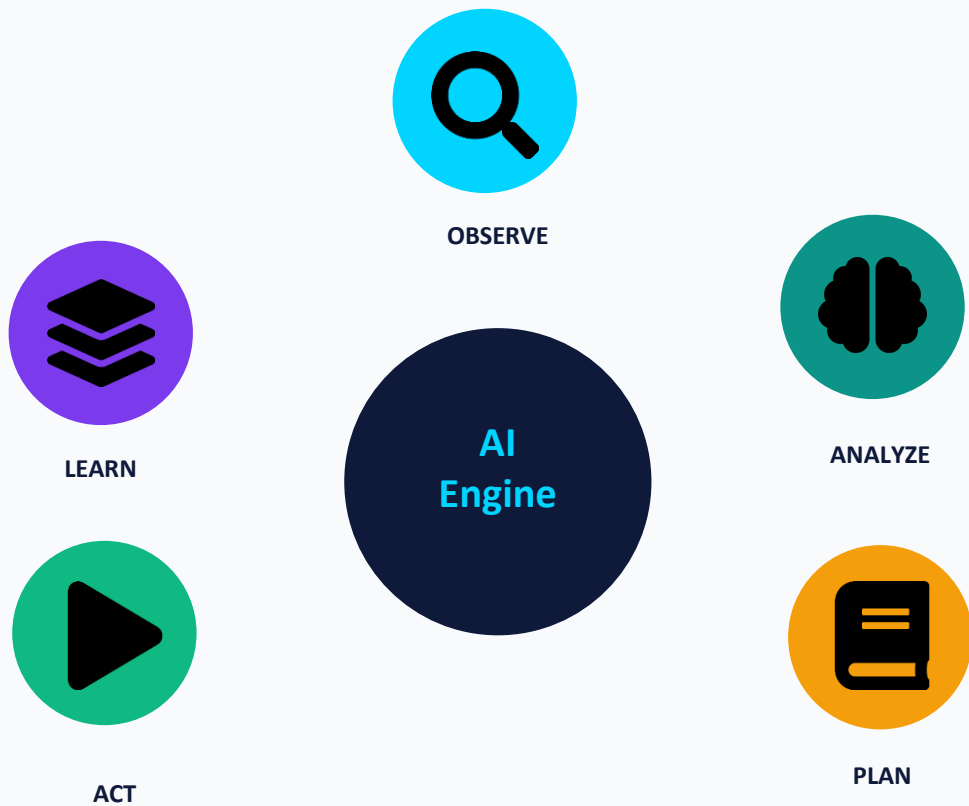
Correct Order: Low → High



↑ LOW AUTONOMY

HIGH AUTONOMY ↑

How AI Thinks: The Decision Cycle



Example

- Food delivery AI
- Detects traffic jam
 - reroutes driver
 - notifies customer
 - updates ETA



What Happens When AI Decides Wrong?

Think about it. Then discuss with the class.



Medical diagnosis



Scheduling



Finance

When AI Gets It Wrong



Immediate Impact

- Wrong output generated
- Wrong action triggered
- Misinterpretation of data
- e.g. Wrong attendance marking



Consequences

- Loss of trust
- Financial damage
- Safety risk
- Algorithmic bias



Root Causes

- Poor training data
- AI hallucination
- Wrong instructions
- Overfitting to patterns

"AI mistakes affect real people — responsible design is not optional."

How Safe AI Systems Protect Us



Human-in-the-Loop

Humans review high-stakes decisions



Self-Correction

AI retries or flags errors automatically



Validation Rules

Checks before taking any action



Logging & Monitoring

Every action is tracked and audited



Fallback Systems

Safe defaults when AI is uncertain



Safe AI systems protect, verify, and always keep humans informed.

Agentic AI: More Power, More Responsibility

The Risk

A mistake by Agentic AI isn't just a wrong answer — it's a wrong action.

Wrong decision made

- Email sent incorrectly
- Profile changed
- ID card issued
- Timetable updated

The Protection



Guardrails

Limit what agent can do



Permissions

Agent must ask for critical actions



Monitoring

Every action is logged & reviewable

Who Is Responsible for AI's Mistakes?

"The AI scheduled the exam on the wrong date."



The AI

made the decision

BUT



Humans

designed, deployed & approved

AI acts. Humans are accountable. Always.

AI Agents Are Already Around You

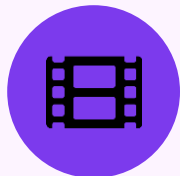
Virtual Assistants



Siri, Alexa, Google Assistant

Respond to voice & control devices

Recommendation Systems



Netflix, YouTube, Spotify

Predict and personalize content

Autonomous Vehicles



Tesla Autopilot, Waymo

Perceive, decide, and navigate

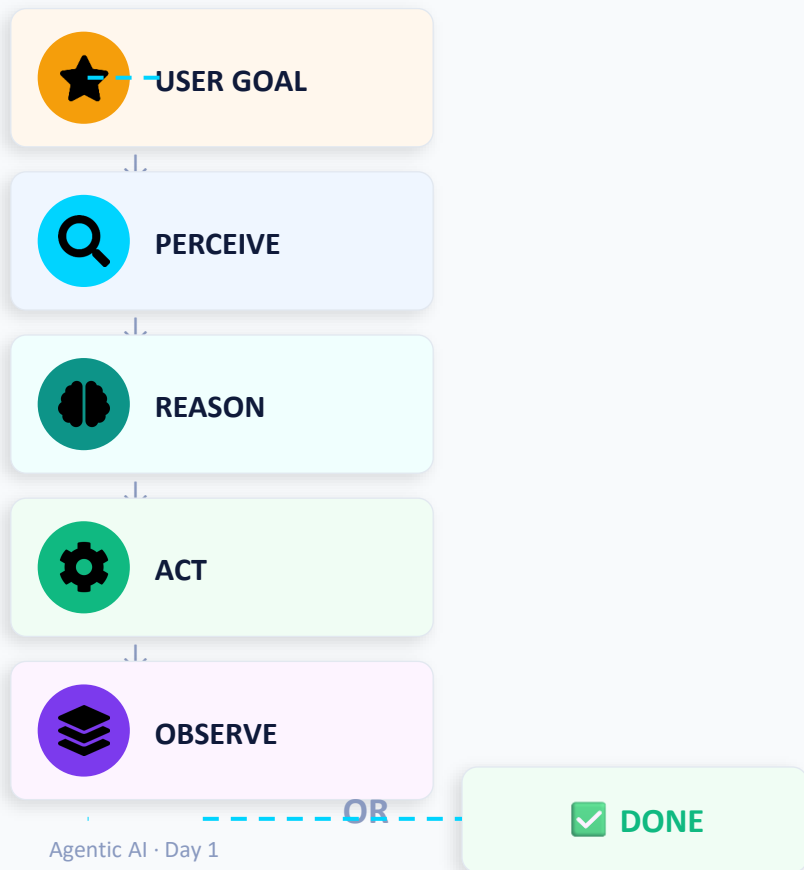
Smart Home Systems



Nest, Philips Hue, Ring

React to environment & routines

The Agent Loop: Perceive → Reason → Act



What's Happening

1. User submits a goal
2. Agent reads & understands context
3. LLM plans the next best step
4. Tools called: APIs, code, email
5. Result observed — done or loop?

⚡ This loop can run 10–100+ times!

Gen AI vs Agentic AI: The Key Differences



Generative AI

Single response per prompt

No memory across calls

Cannot use external tools

Human drives every step

Text / Image / Code

VS



Agentic AI

Multi-step planning

Persistent context

Tool calling — APIs, code

Autonomous goal loop

Completed tasks & actions

Response Style

Memory

Tools

Control

Output

Agentic AI = The next evolution of intelligent systems

Identify the AI Type — 3 Scenarios

1. *AI creates an image from a text description*

Generative AI ✓

2. *AI automatically schedules and confirms meetings end-to-end*

Agentic AI ✓

3. *AI writes an email draft and sends it after your approval*

Hybrid AI ✓

Key Takeaways

01

AI has evolved from rule-based systems to fully autonomous agents

02

RAG = retrieves and answers; Agentic AI = plans and acts

03

An AI agent perceives, reasons, acts, and iterates in a loop

04

Autonomy levels range from human-controlled to fully autonomous

05

More power = more responsibility — for designers AND users

Next Session



Multi-Agent Systems

Agents working together to achieve complex goals



Agent Frameworks

LangChain, AutoGen, CrewAI deep dive



Real-World Architecture

How production agents are actually built



Build Your First Agent!

Hands-on: code a simple AI agent from scratch



Thank You!